# CHEMISTRY

**SECTION - A**

**Multiple Choice Questions:** This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

**Choose the correct answer:**

1. The correct order of electron gain enthalpy (magnitude only) for group 16 elements is

(1) Te > Se > S > O

(2) S > Se > Te > O

(3) O > S > Se > Te

(4) S > O > Se > Te

**Answer (2)**

**Sol.**

| | O | S | Se | Te |
|---|---|---|---|---|
| $\Delta H_{eg}$ (kJ/mol) | –141 | –200 | –195 | –190 |

2. 100 g 98% by weight $H_2SO_4$ is mixed with 100 g 49% by weight $H_2SO_4$. Mole fraction of $H_2SO_4$ in the solution is

(1) 0.9 (2) 0.1

(3) 0.67 (4) 0.33

**Answer (4)**

**Sol.** (I) Mass of $H_2SO_4$ = 98 g

Mass of $H_2O$ = 2 g

(II) Mass of $H_2SO_4$ = 49 g

Mass of $H_2O$ = 51 g

Total mass of $H_2SO_4$ = 98 + 49 = 147 g

$n_{H_2SO_4} = 1.5$ mol

Mass of $H_2O$ = 53 g

$n_{H_2O} \approx 3$ mol

$$X_{H_2SO_4} = \frac{1.5}{4.5} = \frac{1}{3}$$

3. Correct order of ionisation enthalpy is

(1) F > Cl > $Cl^-$ > $F^-$ (2) $F^-$ > $Cl^-$ > F > Cl

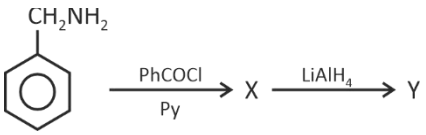
(3) Cl > F > $Cl^-$ > $F^-$ (4) F > Cl > $F^-$ > $Cl^-$

**Answer (1)**

**Sol.**

| | | | | kJ/mol |
|---|---|---|---|---|
| Cl(17) | → | 2, 8, 7 | → | 1256 |
| F(9) | → | 2, 7 | → | 1680 |
| $F^-$(9 + 1) | → | 2, 8 | → | +328 |
| $Cl^-$(17+1) | → | 2, 8, 8 | → | +349 |

4. Consider the following reaction.




The correct structure of Y is

(1) $PhCH_2NHCOPh$

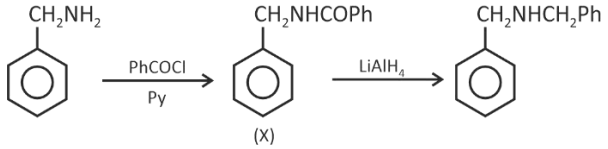
(2) $Ph–CH_2NHCH_2Ph$

(3) $PhNH_2CH_2Ph$

(4) $PhCH_3$



**Answer (2)**

**Sol.** 


5. Which of the following is a mixed oxide?

(1) $Fe_2O_3$ (2) $PbO_2$

(3) $Pb_3O_4$ (4) $BaO_2$

**Answer (3)**

**Sol.** $Pb_3O_4$ is a mixed oxide of PbO and $PbO_2$

6. Which of following is basic buffer?

(1) NaOH + $CH_3COONa$

(2) NaOH + $Na_2SO_4$

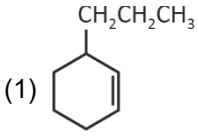
(3) $K_2SO_4$ + $H_2SO_4$

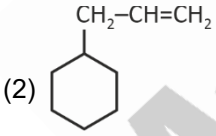
(4) $NH_4OH$ + $NH_4Cl$

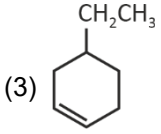
**Answer (4)**

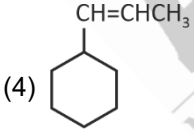
**Sol.** $NH_4OH$ is weak base and $NH_4Cl$ is its salt with strong base

7. An alkene on reductive ozonolysis gives methanal as one of the products. It's structure is

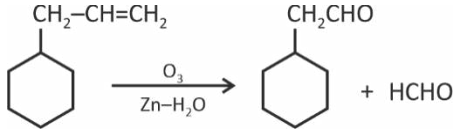
(1) 


(2) 


(3) 


(4) 


**Answer (2)**

**Sol.** 


To get HCHO as one of the product, alkene should contain $=CH_2$ as group.

8. Consider the following data :

$Na(s)+\frac{1}{2}Cl_2(g)\rightarrow NaCl(s)$ $\Delta H^o = -411$ kJ/mole

$Na(s)\rightarrow Na(g)$ $\Delta H^o = 107$ kJ/mole

$Cl_2(g)\rightarrow 2Cl(g)$ $\Delta H^o = 242$ kJ/mole

$Cl(g)+e^-\rightarrow Cl^-(g)$ $\Delta H^o = -355$ kJ/mole

$Na(g)\rightarrow Na^+(g)+e^-$ $\Delta H^o = 502$ kJ/mole$^{-1}$

Find out lattice energy of NaCl(s).

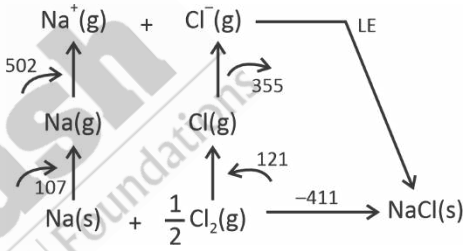
(1) –786 kJ mole$^{-1}$

(2) –628 kJ mol$^{-1}$

(3) –428 kJ mole$^{-1}$

(4) –393 kJ mole$^{-1}$

**Answer (1)**

**Sol.** 


–411 = 107 + 502 + 121 – 355 + LE

LE = –786 kJ mole$^{-1}$

9. Which of the following will behave as best reducing agent?

Given $Al^{3+} + 3e^- \rightarrow Al$ $E° = -1.66$ V

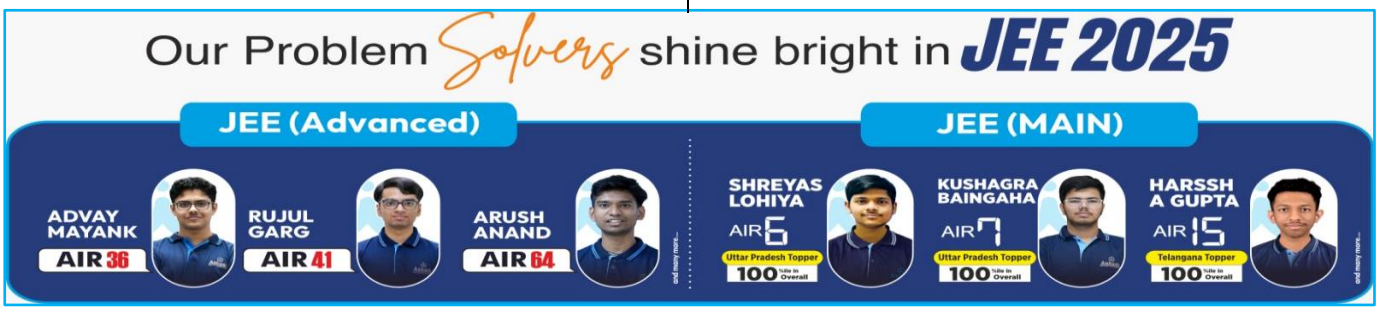
$Co^{3+} + e^- \rightarrow Co^{2+}$ $E° = 1.81$ V

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ $E° = 0.77$ V

$Cd^{2+} + 2e^- \rightarrow Cd$ $E° = -0.4$ V

(1) Al (2) $Co^{2+}$

(3) $Fe^{2+}$ (4) Cd



**Answer (1)**

**Sol.** Smaller the value of E°, lesser will be the tendency to reduce $Al^{3+} + e^- \rightarrow Al$ have smallest E°, so Al will have high tendency to oxidise.

10. $Br_2CH-CH_2-CH_2-CH_3 \xrightarrow[NaNH_2]{2\,Equivalent} X \xrightarrow[II.\,1\,Eq.\,CH_3CHBrCH_3]{I.\,1\,Eq.\,NaNH_2} Y$

(1) 2-Methylhex-3-yne

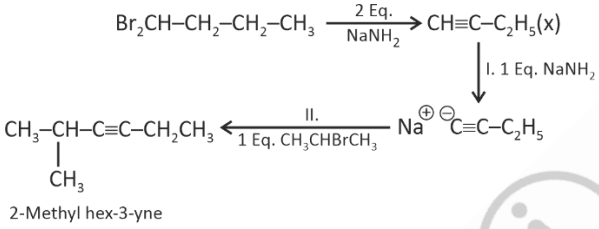
(2) 2-Methylhex-2-ene

(3) 2-Pentyne

(4) 3-Methylhex-2-yne

**Answer (1)**

**Sol.**




2-Methyl hex-3-yne

11. Given below are two statements.

**Statement I :** First ionisation enthalpy of Cr is greater than that of Mn.

**Statement II :** Second and third ionisation enthalpies of Cr are less than that of Mn.

In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is correct but statement II is correct

**Answer (2)**

**Sol.**

| | Cr | Mn |
|---|---|---|
| $IE_1$ (kJ/mol) | 653 | 717 |
| $IE_2$ (kJ/mol) | 1592 | 1509 |
| $IE_3$ (kJ/mol) | 2990 | 3260 |

12. Which of the following is the correct IUPAC name of complex $[Ni(PPh_3)_3(H_2O)_3]Cl_2$ ?

(1) Triaquatris(triphenylphosphine)nickel(II) chloride

(2) Tris(triphenyphosphine)triaquanickel(II) chloride

(3) Triaquatris(triphenylphosphine)nickelate(II) chloride

(4) Triaquatris(triphenylphosphine)nickel(III) chloride

**Answer (1)**

**Sol.** Correct IUPAC name is triaquatris(triphenyl phosphine) nickel(II) chloride

13. Match the two columns : based on the reaction

Glucose + 'X' → 'P'

| | **List-I (Reagent-X)** | | **List-II (Product-P)** |
|---|---|---|---|
| A. | $Br_2$/water | (i) | Glucose oxime |
| B. | Acetic anhydride (excess) | (ii) | Saccharic acid |



| | | | |
|---|---|---|---|
| C. | Conc. $HNO_3$ | (iii) | Glucose pentaacetate |
| D. | $NH_2OH$ | (iv) | Gluconic acid |

(1) A – iv, B – ii, C – iii, D – i

(2) A – ii, B – iv, C – iii, D – i

(3) A – ii, B – iii, C – iv, D – i

(4) A – iv, B – iii, C – ii, D – i

**Answer (4)**

**Sol.** (1) Glucose + $Br_2$-water $\rightarrow$ Gluconic acid

(2) Glucose + Acetic anhydride $\rightarrow$ Glucose pentaacetate

(3) Glucose + Conc. $HNO_3 \rightarrow$ Saccharic acid

(4) Glucose + $NH_2OH \rightarrow$ Glucose oxime

14. Consider two reactions having same pre-exponential factor (A) occuring at same temperature (T).

$A \xrightarrow{E_{a_1}} B$

$C \xrightarrow{E_{a_2}} D$

$E_{a_1} = 5E_{a_2}$

Find out the correct expression?

(1) $\frac{k_1}{k_2} = e^{-\frac{E_{a_2}}{RT}}$ (2) $\frac{k_1}{k_2} = e^{-\frac{4E_{a_1}}{RT}}$

(3) $\frac{k_1}{k_2} = e^{-\frac{4E_{a_1}}{5RT}}$ (4) $\frac{k_1}{k_2} = e^{-\frac{4E_{a_2}}{5RT}}$

**Answer (3)**

**Sol.** $k_1 = Ae^{-\frac{E_{a_1}}{RT}}$

$k_2 = Ae^{-\frac{E_{a_2}}{RT}}$

$\frac{k_1}{k_2} = e^{-\left(\frac{E_{a_1} - E_{a_2}}{RT}\right)}$

$= e^{-\frac{\left(5E_{a_2} - E_{a_2}\right)}{RT}}$

$= e^{-\frac{4E_{a_2}}{RT}}$

$= e^{-\frac{4E_{a_1}}{5RT}}$

Option (3) is correct

15. Consider the given species

$H_2S$, $H_2O$, $NH_3$, $NF_3$, $CHCl_3$

The number of lone pair on central atom which has lowest dipole moment.

(1) 0

(2) 1

(3) 2

(4) 3

**Answer (2)**

**Sol.**

| Molecules | | Dipole Moment (D) |
|---|---|---|
| $NH_3$ | $\rightarrow$ | 1.47 |
| $NF_3$ | $\rightarrow$ | 0.23 |
| $H_2O$ | $\rightarrow$ | 1.85 |
| $H_2S$ | $\rightarrow$ | 0.95 |
| $CHCl_3$ | $\rightarrow$ | 1.04 |



16. Consider the statements below

**Statement I:** $BCl_3$ is covalent in nature

**Statement II:** $BCl_3$ undergoes hydrolysis to form $[B(OH)_4]^-$ and $BH_2^+$

In the light of above statements choose the correct option.

(1) Statement I and statement II both are correct

(2) Statement I and statement II both are incorrect

(3) Statement I correct statement II incorrect

(4) Statement I incorrect statement II correct

**Answer (3)**

**Sol.** $BCl_3$ is covalent in nature

$BCl_3 + H_2O \rightarrow H_3BO_3 + HCl$

17. Energy of first Balmer line of H-atom is x kJ

The energy of second Balmer line of H-atom is ___

(1) x

(2) 1.35 x

(3) 2x

(4) x/1.35

**Answer (2)**

**Sol.** $E_{Balmer\ 1^{st}} \propto 13.6 \times \left(\frac{1}{2^2} - \frac{1}{3^2}\right) \propto 13.6 \times \frac{5}{36} \propto x$ ...(i)

$(n_1 = 2, n_2 = 3)$

$E_{Balmer\ 2^{nd}} \propto 13.6 \times \left(\frac{1}{2^2} - \frac{1}{4^2}\right) \propto 13.6 \times \frac{3}{16}$ ...(ii)

$(ii)/(i) \Rightarrow E_{Balmer\ 2^{nd}} = \frac{13.6 \times \frac{3}{16}}{13.6 \times \frac{5}{36}} \times x = x \times 1.35$

18. 36 g of A reacts with 54 g of B to form $AB_2$, if molar mass of A and B is respectively 60 and 80, then choose correct option from following.

(1) Limiting Reagent is A

(2) 90 g of $AB_2$ formed

(3) Limiting Reagent is B

(4) 50 g of $AB_2$ formed

**Answer (3)**

**Sol.** $n_A = \frac{36}{60} = 0.6,\ n_B = \frac{54}{80} = 0.675$

$$\underset{0.6\,mol}{1A} + \underset{0.675\,mol}{2B} \rightarrow 1AB_2$$

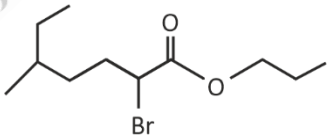
$$\frac{0.6}{1} = 0.6 \qquad \frac{0.675}{2} = 0.3375$$

So, limiting reagent is B

Mole of $AB_2$ formed = 0.3375

W of $AB_2$ = 0.3375 × 220 = 74.25 g

19. The correct IUPAC nomenclature of the following compound is




(1) Propyl-2-bromo-6-methyl heptanoate

(2) 2-Bromo-5-methyl-1-propyl heptanoate

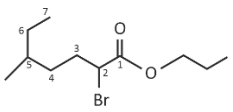
(3) Propyl-2-bromo-5-ethyl hexanoate

(4) Propyl-2-bromo-5-methyl heptanoate

**Answer (4)**



**Sol.** Ester is the principal functional group in this molecule.




→ Propyl-2-bromo-5-methyl heptanoate

20.

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**SECTION - B**

**Numerical Value Type Questions:** This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

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21. Volume ratio of decimolar $NH_4OH$ and decimolar HCl to give a solution of pH = 9.26 at 25°C is x : 1. Find x.

$pK_b$ of $NH_4OH$ = 4.74

**Answer (2)**

**Sol.** Assume $V_{NH_4OH} = xL$ and $V_{HCl} = 1L$

| $NH_4OH$ | + | HCl | → | $NH_4Cl$ | + | $H_2O$ |
|---|---|---|---|---|---|---|
| 0.1x | | 0.1 | | – | | – |
| 0.1x – 0.1 | | 0 | | 0.1 | | |

$$pOH = pK_b + \log\frac{[salt]}{[base]}$$

$$4.74 = 4.74 + \log\frac{(0.1x - 0.1)/V}{0.1/V}$$

0.1x – 0.1 = 0.1

x = 2

22. 5.8 g Aniline is converted into benzanilide with some reaction sequences.

Calculate mass of benzanilide formed, if percentage yield of reaction is 82%.

$$Ph - COCl + PhNH_2 \xrightarrow{Py} \text{Benzanilide}$$

**Answer (10)**

**Sol.** $Ph-\overset{\overset{O}{\|}}{C}-Cl + PhNH_2 \xrightarrow{Py} Ph-\overset{\overset{O}{\|}}{C}-NH-Ph$

Benzanilide

Mole of aniline

$$= \frac{5.8}{93}\text{ mol}$$

Mole of $Ph-\overset{\overset{O}{\|}}{C}-NH-Ph$ will be formed of yield is 100%

$$= \frac{5.8}{93}\text{ mol}$$

mole of $Ph-\overset{\overset{O}{\|}}{C}-NHPh = \frac{5.8}{93} \times 0.82 = 0.0511\text{ mol}$

mass = 10.07 ≈ 10 g

23. A cycloalkene (X) is treated with $Br_2$ and compound (Y) is formed with C : Br ratio 3 : 1. One mole of X required 1 mol of $Br_2$. Find composition of 'Br' in Y compound (percentage).

**Answer (66)**

**Sol.** Cycloalkene (X) $\Rightarrow C_nH_{2n-2}$

After reaction with $Br_2$ compound Y is formed

$\Rightarrow C_nH_{2n-2}Br_2$

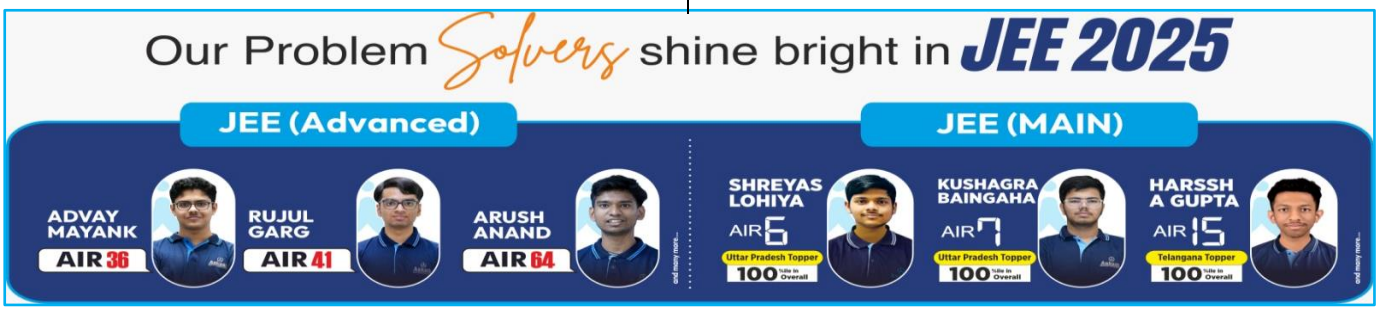
If ration of C : Br = 3 : 1

MF of Y should be $C_6H_{10}Br_2$

$$\%\text{ of Br} = \frac{2\times 80}{242}\times 100 = 66.11\% \approx 66$$

24.

25.



# MATHEMATICS

## SECTION - A

**Multiple Choice Questions:** This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

**Choose the correct answer:**

1. If complex numbers $Z_1, Z_2, \ldots Z_n$ satisfy the equation $4Z^2 + \bar{Z} = 0$, then $\sum_{i=1}^{n} |Z_i|^2$ is equal to

(1) $\frac{3}{16}$ (2) $\frac{3}{64}$

(3) $\frac{9}{64}$ (4) $\frac{1}{16}$

**Answer (1)**

**Sol.** $4z^2 + \bar{z} = 0$

If $z = 0$ then it is a solution

$\Rightarrow |z| = 0$

Now, $z \neq 0$

Multiplying by $z$ we get

$4z^3 + |z|^2 = 0$

$\Rightarrow 4z^3 = -|z|^2$

Taking modulus both sides

$4|z|^3 = |z|^2 \Rightarrow |z|^2(4|z| - 1) = 0$

$\Rightarrow |z| = \frac{1}{4}$

$\Rightarrow 4z^3 = \frac{-1}{16} \Rightarrow z^3 = \frac{-1}{64}$

$\Rightarrow z = \frac{-1}{4}, \frac{-1}{4}w, \frac{-1}{4}w^2$

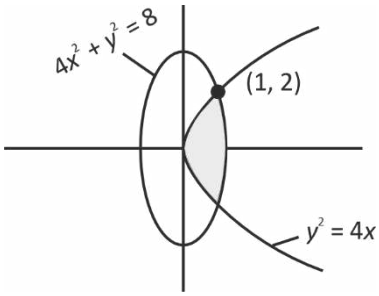
2. Area enclosed by $4x^2 + y^2 \leq 8$ and $y^2 \leq 4x$ is (in square unit) is

(1) $\left(\pi + \frac{4}{3}\right)$ sq. unit (2) $\left(\pi - \frac{4}{3}\right)$ sq. unit

(3) $\left(\pi + \frac{2}{3}\right)$ sq. unit (4) $\left(\pi - \frac{2}{3}\right)$ sq. unit

**Answer (3)**

**Sol.**




$$\text{Area} = 2\int_0^2 \sqrt{\frac{8-y^2}{4}} - \frac{y^2}{4}\,dx$$

$$= \int_0^2 \left(\sqrt{8-y^2} - \frac{y^2}{2}\right)dy$$

$$= 4\sin^{-1}\left(\frac{y}{2^{3/2}}\right) + \frac{y\sqrt{8-y^2}}{2} - \frac{y^3}{6}\Bigg]_0^1$$

$$= 4\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \frac{2\times 2}{2} - \frac{8}{6}$$

$$= \frac{4\pi}{4} + 2 - \frac{4}{3}$$

$$= \left(\pi + \frac{2}{3}\right) \text{sq. unit}$$

3. Mean deviation about median for $k$, $2k$, $3k$,……, $1000k$ is 500, then the value of $k^2$ is

(1) 4

(2) 9

(3) 16

(4) 1

**Answer (1)**

**Sol.** data set

$k$, $2k$, $3k$, ....., $1000k$.

$$\text{Median} = \frac{500k + 501k}{2} = 500.5k$$

Mean-deviation about median

